

Real-Data Machine Learning for Telemedicine Triage and Outbreak Surveillance

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Public real-world data only · no synthetic records · released for exact reproduction



IEEE 6-page paper, 13 references



Seed-42 byte-identical reruns



Fail-closed data pipeline

Two decisions, asymmetric errors

1 Patient-level triage

Which callers can be managed remotely, and which need urgent escalation? Errors are asymmetric: a missed high-acuity patient is a safety event, while an unnecessary referral costs access and trust.

Constraint: 15.2% of encounters are high-acuity; uncertain cases must be deferred, not guessed.

2 Population-level outbreak signal

Is this district-week part of a developing wave? In routine public surveillance there is no outbreak label to train on — only a fixed weekly review budget, so every alert displaces another.

Constraint: a top-5% alert budget, and evaluation that must stay descriptive.



Both layers run on ordinary public data — no proprietary EHR, no private sensor feed, no synthetic records.

Three real public datasets — nothing synthetic

13,595

ED analysis encounters

CDC NHAMCS 2019 public microdata (19,481 raw, protocol exclusions applied) [1]

126,712

district-weeks of SARI signals

covid19india public archive, 840 districts, 2020–2023 [2]

69

months of pharmacy sales

Kaggle monthly series, CC BY-NC 4.0 — supplementary forecast task [3]



Missing raw file ⇒ hard FileNotFoundError



Seed 42 — byte-identical reruns

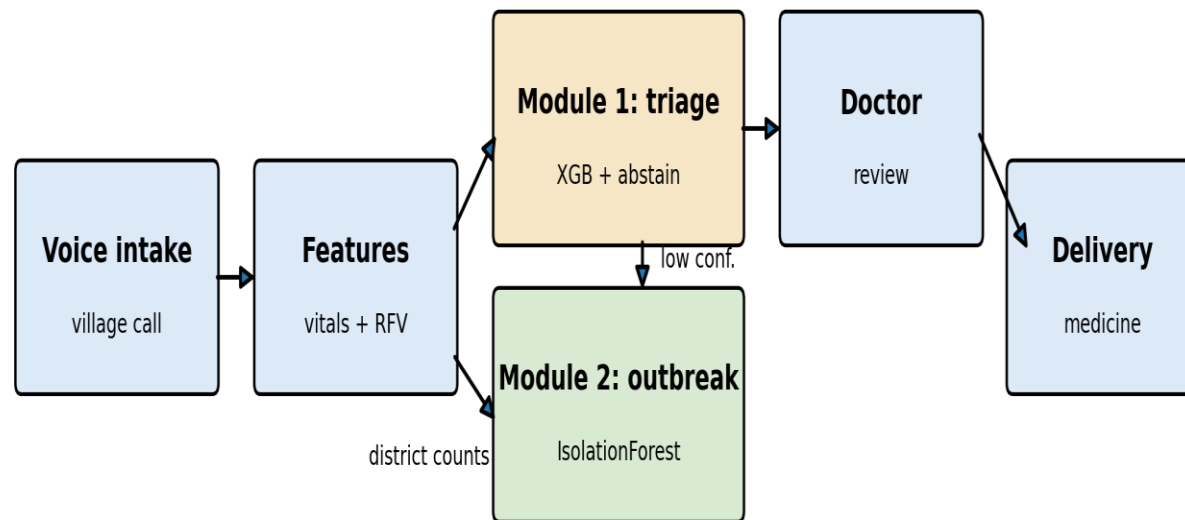


0 synthetic rows anywhere in the pipeline

Every dataset is public and licensed; URLs, retrieval dates and licences are cited in the paper — see Data Availability.

Two modules, one reproducible pipeline

Curago data-science layer: patient-level triage + population-level surveillance



Data: CDC NHAMCS 2019 (public domain) | covid19india archive (static) | Kaggle pharma sales (CC BY-NC, suppl.)

1 Module 1 — patient triage

Calibrated XGBoost (isotonic) risk score → cross-fitted tuned threshold → abstention for uncertain cases → PAC split-conformal recall control at $\alpha = 0.15$.

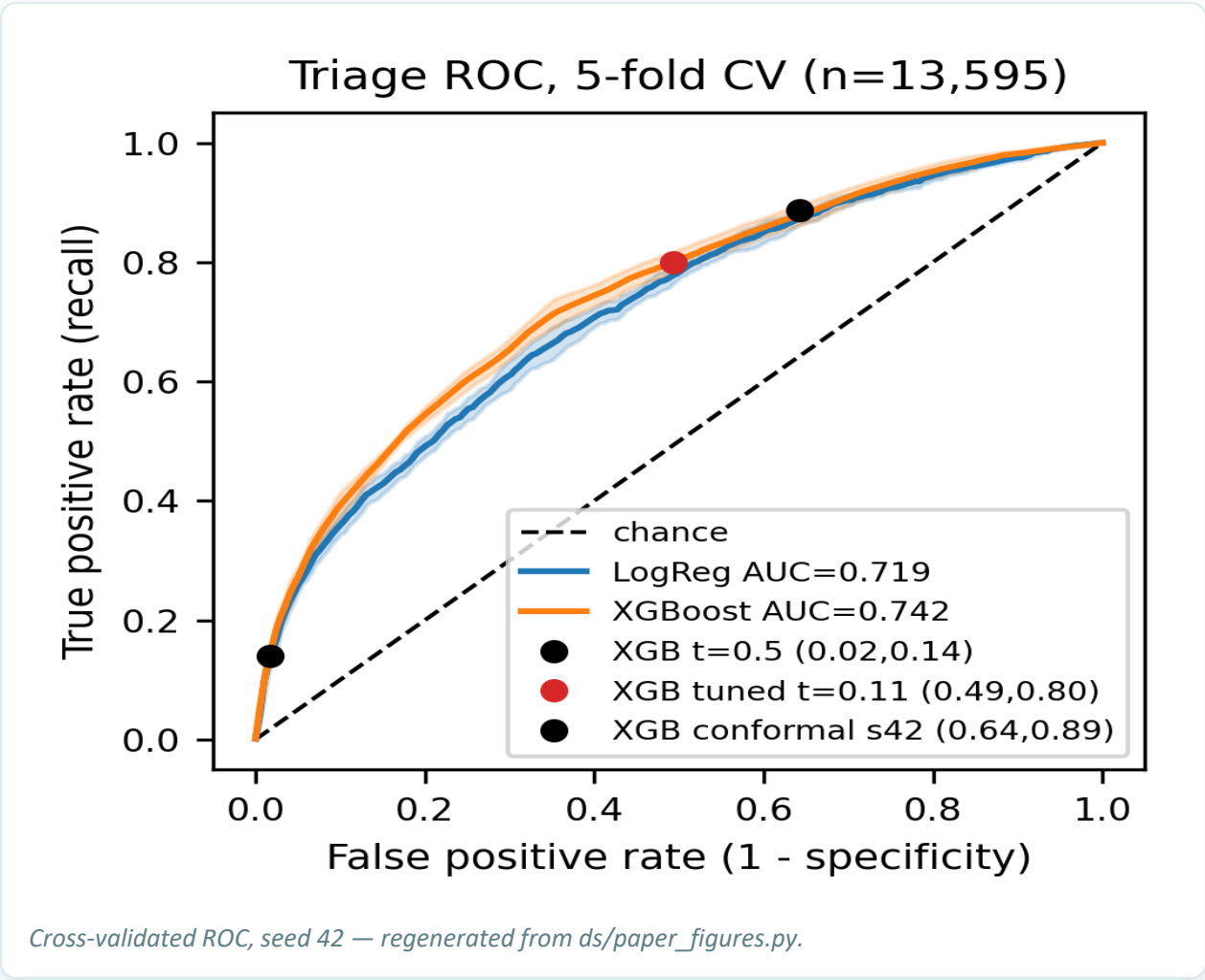
2 Module 2 — district outbreak signal

IsolationForest anomaly score per district-week → fixed 5% weekly alert budget → top-scoring district-weeks surfaced for review, then described retrospectively.



Both modules recompute from raw CSVs; every table and figure in the paper is a generated artifact — never hand-edited.

Module 1 — discrimination and operating point



0.742

primary model CV AUC

vs 0.719 logistic; 95% CI [0.728, 0.751]

17.2%

under-triage, cross-fitted

down from 85.9% at default $t = 0.50$; FPR 0.545

0.829

recall at the chosen point

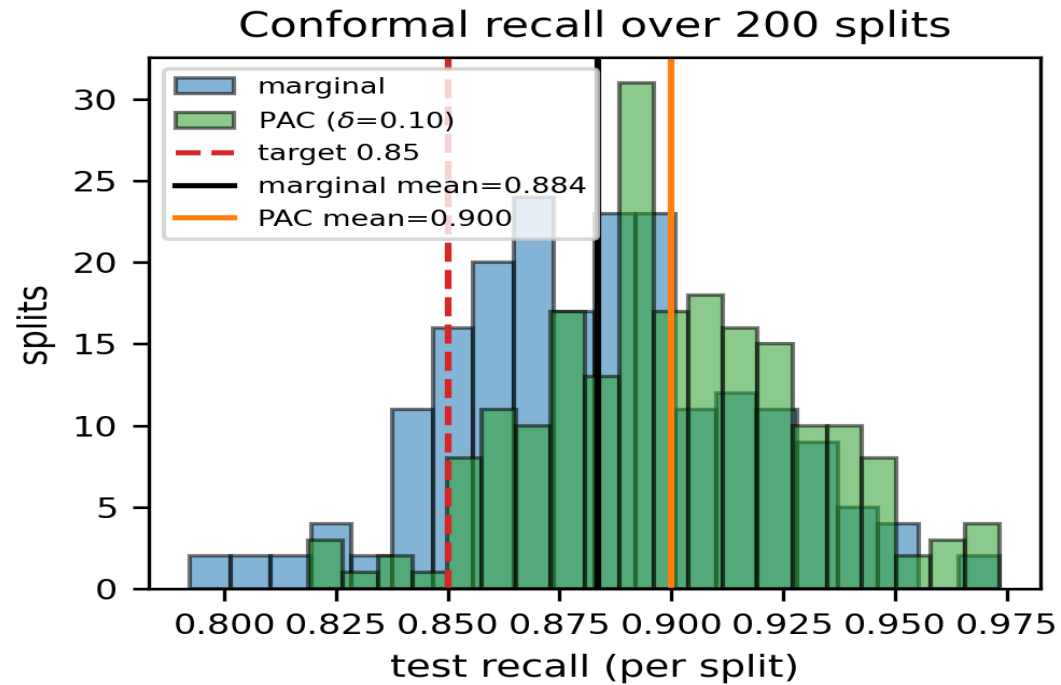
precision 0.214; thresholds 0.09–0.10 per outer fold

0.113

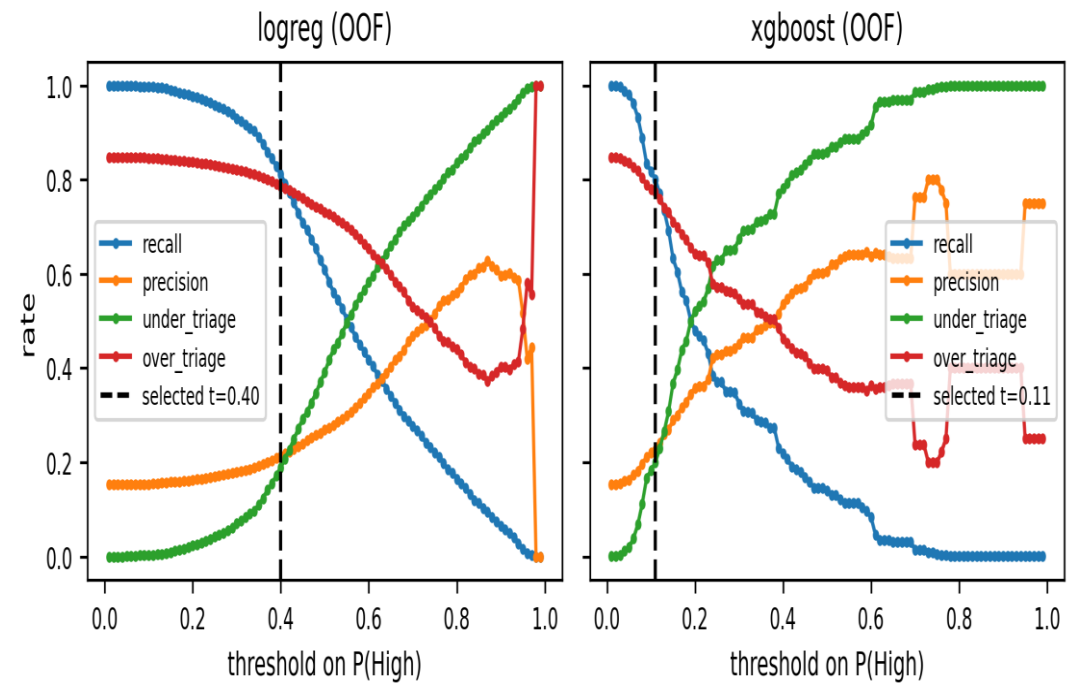
OOF Brier, isotonic

beats 0.129 for a constant — calibration holds

Split-conformal recall control — marginal vs PAC



Recall distribution over 200 random splits, both rules, $\alpha = 0.15$.



Threshold-coverage trade-off, split means $\pm$ SD.



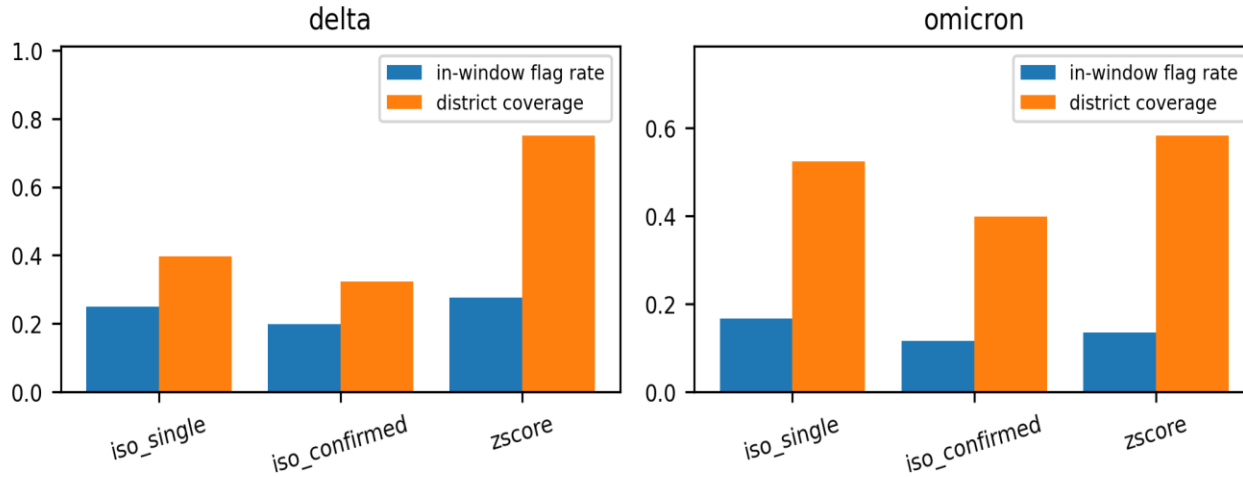
Marginal rule: mean recall 0.884 (SD 0.033), yet 12.5% of splits fall below the 0.85 target — average-case only.



PAC rule ($\delta = 10\%$): mean recall 0.900, below-target splits cut to 3.5% at precision 0.191.

Module 2 — outbreak surveillance, described not claimed

Outbreak methods vs epidemic windows (descriptive)



Weekly IsolationForest flags across 126,712 district-weeks — regenerated from `ds/paper_figures.py`.

7.0×

Delta-wave enrichment

flag rate inside the wave vs background

2 weeks

median lead over the wave

at the fixed 5% alert budget

25%

of Delta weeks flagged

vs 3.6% of weeks outside it



Descriptive only: no labelled outbreak ground truth exists in routine district data [2]. Enrichment is retrospective description at a fixed budget — not a validated detector.

What we do not claim

- ! **Guarantees rest on exchangeability** — The marginal rule is average-case (12.5% of splits below target); PAC bounds that share at $\delta = 10\%$, but neither survives distribution shift.
 - 2 **Transfer is unproven** — Trained on US NHAMCS emergency-department data; rural Indian telemedicine casemix and workflow differ. External validation is future work, not a result.
 - 3 **Calibrated, not oracular** — Isotonic Brier 0.113 beats the constant 0.129 — but probabilities are not validated absolute risks; they drive thresholds, not quotations.
 - 4 **Outbreak detection is descriptive** — IsolationForest enrichment is retrospective description at a 5% budget; naive baselines and NEWS2 stay competitive (cutoff 1: precision 0.168, recall 0.669).
-  All ten mandated disclosures are machine-checked in paper/audit.py and pass — full text in Section V of the paper.

Reproducible by construction



Fail-closed data pipeline

A missing or stale raw file raises `FileNotFoundError`. No silent imputation, no synthetic backfill, no cached numbers on disk.



Seed 42, byte-identical

Repeated runs produce identical CSVs and PNGs; published metrics are read back from generated artifacts at build time.



One JSON behind every number

LaTeX, Markdown, DOCX and this deck all resolve metrics from `paper_numbers.json` — zero hand-typed figures.



Automated audits on every build

Numbers, citations, originality, honesty disclosures and IEEE format are machine-checked — all pass.



github.com/vaibhav7087/curago_paper_version · MIT licence · data licences, URLs and retrieval dates in Data Availability

Takeaways

1

Ordinary public data, tuned without test-fold peeking, beats naive baselines on both tasks — AUC 0.742 for triage (cross-fitted under-triage 17.2%) and 7.0× outbreak enrichment — with protocol limits stated up front.

2

Guarantees must be priced: the marginal conformal rule leaves 12.5% of splits below target, and the PAC rule cuts that to 3.5% at $\delta = 10\%$ — publish the failures alongside the mean.

3

Reproducibility is a feature, not a footnote: fail-closed inputs, byte-identical seed-42 reruns, and one JSON behind every number in four output formats.

Next: prospective validation on rural-Indian telemedicine casemix, drift monitoring, threshold re-tuning — SHAP, calibration and forecast artifacts ship with the repository.

References

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